

Nkosinathi Khumalo
2805324
Campus Marketplace

During this sprint my primary focus was on the backend listing management features, middleware implementation and testing. I implemented the `verifySession`, `attachProfile` and `requireRole` functions in the middleware, ensuring they were correctly wired into the listing management routes so that only authenticated students can modify listings. I expanded on the tests to validate listings and also middleware testing.

One architectural decision I made was to make sure that the system verifies who the seller is automatically, instead of trusting the information sent in the request, meaning that the seller's identity is taken from the logged-in user's session, not from data that could be manipulated. As a result, users cannot pretend to be another student when submitting requests, which fixes a security risk we identified earlier.

CHALLENGES I FACED:

I had to do a lot of debugging when it came to the tests as I am still adjusting to writing unit tests and also connecting the middleware to the routes so that authentication is properly set.

WHAT I WANT TO CHANGE FOR THE NEXT SPRINT:

Writing tests at the start of each feature, rather than at the end. Committing more and ensuring proper synchronization with my team members.

COMMIT HISTORY:

```

Nkosinathi@Nathi-s-Laptop MINGW64 ~/Desktop/campus-marketplace (main)
$ git log --author="Nathi006"
commit 9e0a3ffe515cec700c4ac6155f11682b1aaf00a6 (origin/feature/listing, feature/listing)
Merge: de1f63f 2391d32
Author: Nathi006 <khumalogiven00@gmail.com>
commit 9e0a3ffe515cec700c4ac6155f11682b1aaf00a6 (origin/feature/listing, feature/listing)
Merge: de1f63f 2391d32
Author: Nathi006 <khumalogiven00@gmail.com>
Date: Sat Apr 18 21:42:56 2026 +0200

    Merge branch 'main' into feature/listing

commit de1f63fe911b1a38d59dd143a60ce0d7a3529434
Author: Nathi006 <khumalogiven00@gmail.com>
Date: Sat Apr 18 21:20:16 2026 +0200

    Added validate listing middleware and more tests :)

commit 45d8648f9fe875e675341bfc0ec524fe70f29fd8
commit 9e0a3ffe515cec700c4ac6155f11682b1aaf00a6 (origin/feature/listing, feature/listing)
Merge: de1f63f 2391d32
Author: Nathi006 <khumalogiven00@gmail.com>
Date: Sat Apr 18 21:42:56 2026 +0200

    Merge branch 'main' into feature/listing

commit de1f63fe911b1a38d59dd143a60ce0d7a3529434
Author: Nathi006 <khumalogiven00@gmail.com>
Date: Sat Apr 18 21:20:16 2026 +0200

    Added validate listing middleware and more tests :)

commit 45d8648f9fe875e675341bfc0ec524fe70f29fd8
Author: Nathi006 <khumalogiven00@gmail.com>
Date: Sat Apr 18 01:11:53 2026 +0200

    Added unit testing for middleware & data integration

```